Communication Systems - Lecture Notes

1. Amplitude Modulation

In Amplitude Modulation (AM) the carrier amplitude varies with the message: s(t) = Ac*(1 + m*cos(2*pi*fm*t))*cos(2*pi*fc*t), where m is the modulation index. For distortion-free demodulation m must satisfy 0 < m <= 1. The bandwidth of a standard AM signal is twice the message bandwidth, BW = 2*fm.

The modulation index is m = Am/Ac, the ratio of message amplitude to carrier amplitude. An AM signal contains the carrier and two sidebands; suppressing the carrier gives double-sideband suppressed-carrier (DSB-SC), which requires coherent detection.

2. Angle Modulation

In Frequency Modulation (FM) the instantaneous frequency deviates with the message. The peak frequency deviation is delta_f, and the modulation index is beta = delta_f/fm. Carson's rule approximates the bandwidth as BW = 2*(delta_f + fm), or 2*(beta + 1)*fm.

FM is more robust to amplitude noise than AM because the information is carried in the frequency, but it requires a wider bandwidth.

3. Information Theory

Channel capacity in the presence of additive white Gaussian noise is given by the Shannon-Hartley theorem: C = B*log2(1 + S/N) bits per second, where B is the channel bandwidth in hertz and S/N is the signal-to-noise power ratio (a linear ratio, not in decibels).

The Shannon limit states that error-free communication is possible only for rates below C. Doubling the bandwidth does not double capacity, because adding bandwidth also adds noise power.

4. Digital Modulation

Common digital schemes include Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Phase Shift Keying (PSK). Binary PSK (BPSK) uses two phases 180 degrees apart. Quadrature PSK (QPSK) carries two bits per symbol using four phases. Quadrature Amplitude Modulation (QAM) combines amplitude and phase; 16-QAM carries 4 bits per symbol and 64-QAM carries 6 bits per symbol.